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Power tool and battery pack

Abstract

A power tool includes an output member for outputting power, a motor for driving the output member, a housing including a battery pack accommodating portion provided with a first battery pack accommodating cavity and a guide structure for guiding a battery pack to be inserted into the first battery pack accommodating cavity along a first straight line. The battery pack includes a basic housing portion provided with a first accommodating cavity, a first cell group disposed inside the first accommodating cavity, an extended housing portion provided with a second accommodating cavity, and a second cell group disposed inside the second accommodating cavity. The first cell group is at least partially disposed inside the first battery pack accommodating cavity, and the second cell group is at least partially disposed outside the first battery pack accommodating cavity.

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Background/Summary

RELATED APPLICATION INFORMATION

(1) This application claims the benefit under 35 U.S.C. § 119(a) of Chinese Patent Application No. CN 202111012566.2, filed on Aug. 31, 2021, Chinese Patent Application No. CN 202111014289.9, filed on Aug. 31, 2021, Chinese Patent Application No. CN 202122093651.8, filed on Aug. 31,

2021, Chinese Patent Application No. CN 202111014249.4, filed on Aug. 31, 2021, Chinese Patent Application No. CN 202122091164.8, filed on Aug. 31, 2021, and Chinese Patent Application No. CN 202122089383.2, filed on Aug. 31, 2021, which applications are incorporated herein by reference in their entirety.

BACKGROUND

(2) Power tools on the market are generally powered in two manners. One is an external power supply, and the other is a battery pack. If the external power supply is used, the power tools can only work in places with power interfaces, which greatly limits the range where the power tools are applied. The battery pack can solve the problem well. However, the power tools will work under various working conditions, and the requirement for the endurance of the battery pack depends on different working conditions. In some cases, the power tools are required to work for a long time, and thus a large-capacity battery pack needs to be mounted. In some other cases, the battery pack is not required to have relatively great endurance, and the installation of a relatively large battery pack causes inconvenience to a user. Currently, the power tools on the market cannot solve this problem.

SUMMARY

(3) A power tool includes an output member for outputting power, a motor for driving the output member, a housing for accommodating at least part of the motor, and a battery pack for powering the motor. The housing includes a battery pack accommodating portion provided with a first battery pack accommodating cavity and a guide structure for guiding the battery pack to be inserted into the first battery pack accommodating cavity along a direction of a first straight line. The battery pack includes a basic housing portion provided with a first accommodating cavity, a first cell group disposed inside the first accommodating cavity, an extended housing portion provided with a second accommodating cavity, and a second cell group disposed inside the second accommodating cavity. The first cell group is at least partially disposed inside the first battery pack accommodating cavity, and the second cell group is at least partially disposed outside the first battery pack accommodating cavity.

(4) A battery pack is applicable to a power tool and detachably mounted to a first battery pack accommodating cavity of the power tool. The battery pack includes a battery pack housing, a cell assembly, an interface circuit board, and an electrode plate terminal assembly. The cell assembly includes a first cell group and a second cell group which are disposed inside the battery pack housing, where the first cell group is at least partially disposed inside the first battery pack accommodating cavity, and the second cell group is at least partially disposed outside the first battery pack accommodating cavity. The interface circuit board is at least partially disposed inside the battery pack housing. The electrode plate terminal assembly is used for electrically connecting the battery pack to tool terminals of the power tool, where electrical conductivity of the electrode plate terminal assembly is greater than or equal to 30% IACS and less than or equal to 100% IACS, and the electrode plate terminal assembly includes at least a positive terminal assembly and a negative terminal assembly which are mounted to the interface circuit board.

(5) A battery pack is applicable to a power tool and detachably mounted to a first battery pack accommodating cavity of the power tool. The battery pack includes a battery pack housing and an interface circuit board which is mounted to the battery pack housing. The battery pack further includes a first cell group, a second cell group, and a connection device. The first cell group is disposed inside the battery pack housing, where the first cell group includes multiple first cells and a first positive terminal and a first negative terminal which are used for electrically connecting the first cell group to the interface circuit board. The second cell group is disposed inside the battery pack housing, where the second cell group includes multiple second cells and a second positive terminal and a second negative terminal which are used for electrically connecting the second cell group to the interface circuit board, the first cell group is at least partially disposed inside the first battery pack accommodating cavity, and the second cell group is at least partially disposed outside

the first battery pack accommodating cavity. The connection device is electrically connected to the first positive terminal and the second positive terminal, where the connection device is electrically connected to the interface circuit board. Alternatively, the connection device is electrically connected to the first negative terminal and the second negative terminal, where the connection device is electrically connected to the interface circuit board.

Description

BRIEF DESCRIPTION OF DRAWINGS
(1) FIG. **1** is a perspective view of a blower shown in the present application;
(2) FIG. **2** is a sectional view of the blower shown in FIG. **1**;
(3) FIG. **3** is a perspective view of a tool body and a battery pack of the blower shown in FIG. **1**, which are separated from each other;
(4) FIG. **4** is an exploded view of part of a tool body of the blower shown in FIG. **1**;
(5) FIG. **5** is an exploded perspective of part of the tool body of the blower shown in FIG. **4** from another angle;
(6) FIG. **6** is an exploded view of part of a tool body of the blower shown in FIG. **1**;
(7) FIG. **7** is a perspective view of a battery pack of the blower shown in FIG. **1**;
(8) FIG. **8** is a perspective view of the battery pack shown in FIG. **7** from another angle;
(9) FIG. **9** is a plan view of the battery pack shown in FIG. **7**;
(10) FIG. **10** is a plan view of the battery pack shown in FIG. **7**;
(11) FIG. **11** is a perspective view of part of the battery pack shown in FIG. **7**;
(12) FIG. **12** is a perspective view of the part of the battery pack shown in FIG. **11** from another angle;
(13) FIG. **13** is a plan view of the part of the battery pack shown in FIG. **11**;
(14) FIG. **14** is a perspective view of an arrangement manner of a cell assembly in the battery pack shown in FIG. **11**;
(15) FIG. **15** is an exploded view of the arrangement manner of the cell assembly in the battery pack shown in FIG. **14**;
(16) FIG. **16** is a perspective view of another arrangement manner of a cell assembly in the battery pack shown in FIG. **14**; and
(17) FIG. **17** is an exploded view of the arrangement manner of the cell assembly in the battery pack shown in FIG. **16**.
DETAILED DESCRIPTION
(18) As shown in FIG. **1**, a power tool for mounting a battery pack **40** to a tool body **100***a* along a direction of a first straight line **101** is provided, which is specifically exemplified by a blower **100** in the present application. The blower **100** can be operated by a user to clean things such as leaves, grass, snow, and dust. Of course, it is to be understood that the power tool may be another type of tool such as a string trimmer, a snow thrower, a pole saw, or a chain saw. In fact, as long as these tools include the substance described below in the present application, these tools all fall within the scope of the present application.
(19) As shown in FIGS. **1** to **3**, the blower **100** includes the tool body **100***a* and the battery pack **40**. The tool body **100***a* includes an output member **10**, a motor **20**, a housing **30**, a handle **70**, and a switch assembly **80**.
(20) The output member **10** is a fan **10***a* and used for outputting power, that is, the fan **10***a* rotates to generate an airflow.
(21) The motor **20** is used for driving the fan **10***a* to rotate.
(22) An accommodating space **37** is formed in the housing **30**, and the motor **20** is at least partially disposed in the accommodating space **37**. The housing **30** is provided with or fixedly connected to

an air tube **38**, and the air tube **38** is provided with an air duct through which the airflow circulates. That is, the motor **20** drives the fan **10***a* to rotate, thereby generating the airflow which is finally blown out from the air tube **38**. The handle **70** for the user to hold is fixedly connected to the housing **30** or integrally formed with the housing **30**. The switch assembly **80** for starting the motor **20** is disposed on the handle **70** such that the user can operate the switch assembly **80** conveniently when holding the handle **70**.

(23) The battery pack **40** is used for powering the motor **20**. The battery pack **40** can be mounted to the housing **30** along the direction of the first straight line **101**.

(24) As shown in FIGS. **7** to **10**, the battery pack **40** includes a basic portion **41**, an extended portion **43**, and a connecting portion **42**, where the basic portion **41** is spaced apart from the extended portion **43**, and the connecting portion **42** is disposed between the basic portion **41** and the extended portion **43** and used for connecting the basic portion **41** to the extended portion **43**. The extended portion **43** is connected in parallel to the basic portion **41** so that the capacity of the battery pack **40** is extended, that is, the capacity of the extended portion **43** may be increased as required so as to increase the total capacity of the battery pack **40**.

(25) A battery pack housing **47** forms the appearance of the battery pack **40**. The battery pack housing **47** includes a basic housing portion **44**, a connecting housing portion **421**, and an extended housing portion **431**, where the basic housing portion **44** is spaced apart from the extended housing portion **431**, and the connecting housing portion **421** is disposed between the basic housing portion **44** and the extended housing portion **431** and used for connecting the basic housing portion **44** to the extended housing portion **431**. That is, the connecting housing portion **421** is disposed between the basic housing portion **44** and the extended housing portion **431** so as to form a gap **60** between the basic housing portion **44** and the extended housing portion **431**, where the gap **60** is outside the battery pack housing **47**. When the battery pack **40** is mounted to the housing **30**, the basic housing portion **44** and the extended housing portion **431** are disposed on two sides of the housing **30** and the housing **30** at least partially passes through the preceding gap **60**. The battery pack **40** powers the blower **100** for a long time, and a large amount of heat is necessarily generated inside the battery pack **40**, which will affect the service lives of internal components of the battery pack **40** in the long run. The basic housing portion **44** and the extended housing portion **431** are spaced apart from each other so that a heat dissipation area of the battery pack **40** can be increased, thereby facilitating the heat dissipation of the battery pack **40**.

(26) Referring to FIGS. **11** and **13**, additionally, the battery pack **40** further includes a support **472**, a cell assembly **471**, an electrode plate terminal assembly **481**, an interface circuit board **48**, and a connection device **49**. The support **472** is fixedly connected to or directly formed by the battery pack housing **47**. The support **472** is used for supporting the cell assembly **471**. It may also be understood as follows: the cell assembly **471** is fixed on the support **472**, and further the cell assembly **471** is fixed in the battery pack housing **47**. The cell assembly **471** includes a first cell group **45** and a second cell group **46**, both of which are used for storing power. The interface circuit board **48** is fixedly mounted in the battery pack housing **47**. The interface circuit board **48** is disposed at an end of the cell assembly **471** along an insertion direction of the tool terminals **361**. The interface circuit board **48** is used for electrically connecting the first cell group **45** and the second cell group **46** to the electrode plate terminal assembly **481**. The electrode plate terminal assembly **481** is mounted onto the interface circuit board **48** and used for electrically connecting the battery pack **40** to the tool terminals **361** of an external tool. That is, it may be understood as follows: the electrode plate terminal assembly **481** is used for connecting the tool terminals **361** on the tool body **100***a*, thereby supplying the power stored in the first cell group **45** and the second cell group **46** to the tool body **100***a*. Of course, the electrode plate terminal assembly **481** may also be used for connecting a charger to charge the first cell group **45** and the second cell group **46**. The connection device **49** is electrically connected to the interface circuit board **48**, where the connection device **49** and the interface circuit board **48** may be connected to each other through a

connection piece, a wire, or the like having a conductive property. In this example, the connection device **49** and the interface circuit board **48** may be connected to each other through a wire **491** such that the arrangement of the internal structures of the battery pack **40** is more proper.
(27) The first cell group **45** is disposed inside a first accommodating cavity **411** formed by the basic housing portion **44**, and the second cell group **46** is disposed inside a second accommodating cavity **432** formed by the extended housing portion **431**. A projection of the first cell group **45** in a plane C perpendicular to the first straight line **101** does not overlap a projection of the second cell group **46** in the plane C. That is, it may be understood as follows: the first cell group **45** and the second cell group **46** do not coincide with each other and the second cell group **46** is disposed at the upper side or the lower side of the first cell group **45** along an up-down direction parallel to the plane C.
(28) The first cell group **45** includes multiple first cells **451**, and the second cell group **46** includes multiple second cells **461**, where the first cells **451** are disposed inside a first battery pack accommodating cavity **32** and the second cells **461** are disposed inside a second battery pack accommodating cavity **33** when the battery pack **40** is mounted to the housing **30**. As an example, the first cell unit **451** is a cylinder centered on a first center line **103**, and the second cell unit **461** is a cylinder centered on a second center line **104**. A direction of the first center line **103** is parallel to a direction of the second center line **104**. It may also be understood as follows: the first cells **451** and the second cells **461** are disposed in parallel in the battery pack housing **47**. The direction of the first center line **103** is parallel to the direction of the first straight line **101**, that is, a direction in which the first cells **451** and the second cells **461** are disposed is consistent with a direction in which the battery pack **40** is inserted into the tool body **100***a*. As another example, of course, the direction of the first center line and the direction of the second center line are perpendicular to each other, and the direction of the first center line is parallel to the direction of the first straight line, that is, a direction in which the first cells are disposed is parallel to the direction in which the battery pack is inserted, and a direction in which the second cells are disposed is perpendicular to the direction in which the battery pack is inserted. As another example, the direction of the first center line and the direction of the second center line are perpendicular to each other, and the direction of the first center line is perpendicular to the direction of the first straight line, that is, the direction in which the first cells are disposed is perpendicular to the direction in which the battery pack is inserted, and the direction in which the second cells are disposed is parallel to the direction in which the battery pack is inserted. As another example, the direction of the first center line and the direction of the second center line are parallel to each other, the direction of the first center line is perpendicular to the direction in which the battery pack is inserted, and the direction of the second center line is perpendicular to the direction in which the battery pack is inserted, that is, the direction in which the first cells are disposed is perpendicular to the direction in which the battery pack is inserted, and the direction in which the second cells are disposed is perpendicular to the direction in which the battery pack is inserted. In this example, preferably, the direction of the first center line **103** is parallel to the direction of the second center line **104**, and the direction of the first center line **103** is parallel to the direction of the first straight line **101**, that is, the direction in which the first cells **451** and the second cells **461** are disposed is parallel to the direction in which the battery pack **40** is inserted, and the first cell group **45** and the second cell group **46** are disposed in parallel in the direction perpendicular to the first straight line **101**.
(29) The first cell group **45** includes the multiple first cells **451** disposed in a first plane P**1**, and the second cell group **46** includes the multiple second cells **461** disposed in a second plane P**2**, where the first plane P**1** and the second plane P**2** are parallel to each other. It is to be noted that the first cell group **45** and the second cell group **46** include at least multiple first cells **451** and multiple second cells **461** respectively along the direction perpendicular to the first straight line **101**. For the sake of clarity, it is now defined that along the direction perpendicular to the first straight line **101**, the first cell unit **451** and the second cell unit **461** with a shortest distance between the first center line **103** and the second center line **104** are located in the first plane P**1** and the second plane P**2**,

respectively. That is, along the direction perpendicular to the first straight line **101**, the first cell unit **451** and the second cell unit **461** with the shortest distance have centers located in the first plane P**1** and the second plane P**2**. A distance between the first plane P**1** and the second plane P**2** is greater than or equal to 30 mm and less than or equal to 80 mm. The distance between the first plane P**1** and the second plane P**2** is set within the preceding proper range so that the electrical connection between the first cell group **45** and the second cell group **46** can be more reliable, and assembly and manufacturing difficulties are reduced.

(30) Further, the first cell group **45** at least includes three or more first cells **451** disposed in the first plane P**1**, and the second cell group **46** at least includes three or more second cells **461** disposed in the second plane P**2**, where the first plane P**1** and the first straight line **103** are parallel to each other or coincide with each other. Further, a ratio of the number of the first cells **451** to the number of the second cells **461** is greater than or equal to 0.3 and less than or equal to 2. That is, the user may arrange different numbers of cell units as required so that extended portions **43** with different capacities can be selected. Thus, the user has more choices. To prevent the battery pack **40** from being too long in a thickness direction, the multiple first cells **451** of the first cell group **45** may be arranged along the direction of the first center line **103**. Multiple first cells **451** may be disposed in a direction perpendicular to the first center line **103**. Preferably, two first cells **451** may be arranged along the direction of the first center line **103** such that the basic portion **41** can be prevented from being too long in a length direction and the thickness direction.

(31) As an example, as shown in FIGS. **14** and **15**, a first positive terminal **452** and a first negative terminal **453** which are used for electrically connecting the first cell group **45** to the interface circuit board **48** are mounted on the first cell group **45**. Similarly, a second positive terminal **462** and a second negative terminal **463** which are used for electrically connecting the second cell group **46** to the interface circuit board **48** are mounted on the second cell group **46**. The connection device **49** electrically connects the first positive terminal **452** to the second positive terminal **462**, thereby electrically connecting the first cell group **45** to the second cell group **46**. In this example, the connection device **49** is disposed in the battery pack housing **47**, and the interface circuit board **48** and at least part of the connection device **49** are disposed opposite to each other on two sides of the first cell group **45**. It may also be understood as follows: the interface circuit board **48** and at least part of the connection device **49** are disposed at two ends of the inner space of the battery pack housing **47**, separately. Specifically, the connection device **49** includes a connection plate **492** and a long electrode plate **493**. The connection plate **492** is disposed at one end of the first cell group **45** and the interface circuit board **48** is disposed at the other end of the first cell group **45**. Further, the connection plate **492** is connected to the first positive terminal **452** through a metal connection piece **494**, and the connection plate **492** is connected to the second positive terminal **462** through a metal connection piece **494**. The long electrode plate **493** electrically connects the first cell group **45** to the second cell group **46** so that the first cell group **45** and the second cell group **46** are connected in parallel. The connection plate **492** and the interface circuit board **48** are connected to each other through the wire **491** such that positive terminals of the first cell group **45** and the second cell group **46** can be connected to the interface circuit board **48** simply through one circuit, thereby simplifying a connection structure in the battery pack **40**. Along the direction of the first center line **103**, the long electrode plate **493** is disposed in the middle of the first cell group **45**, that is, the long electrode plate **493** is not disposed at two ends of the first cell group **45**. The long electrode plate **493** may be substantially disposed between first cells **451** and close to the second cell unit **461**, that is, the long electrode plate **493** is disposed between the first cell unit **451** and the second cell unit **461** in the direction perpendicular to the first center line **103**.

(32) As another example, as shown in FIGS. **16** and **17**, a first positive terminal **452***a* and a first negative terminal **453***a* which are used for electrically connecting a first cell group **45***a* to an interface circuit board **48***a* are mounted on the first cell group **45***a*. Similarly, a second positive terminal **462***a* and a second negative terminal **463***a* which are used for electrically connecting a

second cell group **46***a* to the interface circuit board **48***a* are mounted on the second cell group **46***a*. A connection device **49***a* electrically connects the first negative terminal **453***a* to the second negative terminal **463***a*, thereby electrically connecting the first cell group **45***a* to the second cell group **46***a*. In this example, the connection device **49***a* is disposed in the battery pack housing, and the connection device **49***a* and the interface circuit board **48***a* are disposed opposite to each other on two sides of the first cell group **45***a*. It may also be understood as follows: the connection device **49***a* and the interface circuit board **48***a* are disposed at two ends of the inner space of the battery pack housing, separately. The connection device **49***a* includes a connection plate **492***a* and a long electrode plate **493***a*. The connection plate **492***a* is disposed at one end of the first cell group **45***a* and the interface circuit board **48***a* is disposed at the other end of the first cell group **45***a*. Further, the connection plate **492***a* is connected to the first negative terminal **453***a* through a metal connection piece **494***a*, and the connection plate **492***a* is connected to the second negative terminal **463***a* through a metal connection piece **494***a*. The long electrode plate **493***a* electrically connects the first cell group **45***a* to the second cell group **46***a* so that the first cell group **45***a* and the second cell group **46***a* are connected in parallel. The connection plate **492***a* and the interface circuit board **48***a* are connected to each other through a wire **491***a* such that negative terminals of the first cell group **45***a* and the second cell group **46***a* can be connected to the interface circuit board **48***a* simply through one circuit, thereby simplifying a connection structure in the battery pack. Along a direction of a first center line **103***a*, the long electrode plate **493***a* is disposed in the middle of the first battery cell group **45***a*, that is, the long electrode plate **493***a* is not disposed at two ends of the first cell group **45***a*. The long electrode plate **493***a* may be substantially disposed between first cells **451***a* and close to the second cell unit **461***a*, that is, the long electrode plate **493** is disposed between the first cell unit **451***a* and the second cell unit **461***a* in a direction perpendicular to the first center line **103***a*.

(33) As shown in FIGS. **11** to **13**, the electrode plate terminal assembly **481** is at least partially mounted inside the battery pack housing **47**. Further, the electrode plate terminal assembly **481** is at least partially disposed inside the basic housing portion **44**. The electrode plate terminal assembly **481** is used for supplying the power of the first cell group **45** and the second cell group **46** to the motor **20** in the tool body **100***a*. In this example, electrical conductivity of the electrode plate terminal assembly **481** is greater than or equal to 30% IACS and less than or equal to 100% IACS, that is, the electrical conductivity of the electrode plate terminal assembly **481** is set within the preceding range, thereby ensuring the conductive property between the electrode plate terminal assembly **481** and the tool terminals **361**. Further, the electrical conductivity of the electrode plate terminal assembly **481** is greater than or equal to 50% IACS and less than or equal to 90% IACS so that the connection stability and the conductive property between the electrode plate terminal assembly **481** and the tool terminals **361** can be ensured.

(34) The electrode plate terminal assembly **481** includes a positive terminal assembly **482**, a negative terminal assembly **483**, and a signal terminal assembly **484** which are mounted on the interface circuit board **48**. The positive terminal assembly **482**, the negative terminal assembly **483**, and the signal terminal assembly **484** are fixedly connected to the interface circuit board **48**. The positive terminal assembly **482** is connected to a positive electrode of the cell assembly **471**, and the negative terminal assembly **483** is connected to a negative electrode of the cell assembly **471**. The positive terminal assembly **482** and the negative terminal assembly **483** are in contact with the tool terminals **361** on the tool body **100***a* so that the battery pack **40** is electrically connected to external equipment. The signal terminal assembly **484** is used for connecting a tool communication terminal of the blower **100** so as to establish a communication connection between the battery pack **40** and the blower **100** so that the data communication between the battery pack **40** and the blower **100** can be implemented. Further, the positive terminal assembly **482** is used for connecting a positive connection terminal of the blower **100**, and the negative terminal assembly **483** is used for connecting a negative connection terminal of the blower **100**, where the positive terminal assembly

482 and the negative terminal assembly **483** are used for enabling the battery pack **40** to transmit electric energy to the blower **100**. The positive terminal assembly **482**, the negative terminal assembly **483**, and the signal terminal assembly **484** of the electrode plate terminal assembly **481** on the interface circuit board **48** all extend in the insertion direction of the tool terminals **361**. In addition, the positive terminal assembly **482**, the negative terminal assembly **483**, and the signal terminal assembly **484** of the electrode plate terminal assembly **481** on the interface circuit board **48** are substantially parallel to each other along a direction perpendicular to the insertion direction of the tool terminals **361**. The insertion direction of the tool terminals **361** is substantially parallel to the direction of the first straight line **101**. A dimension F**1** of the positive terminal assembly **482** along the insertion direction of the tool terminals **361** is larger than or equal to 7 mm and smaller than or equal to 15 mm. The dimension of the positive terminal assembly **482** is set within the preceding range, which can ensure the connection stability between the positive terminal assembly **482** and the tool terminals **361** and will not cause too long a dimension of the battery pack **40** in the length direction. A dimension F**2** of the negative terminal assembly **483** along the insertion direction of the tool terminals **361** is larger than or equal to 7 mm and smaller than or equal to 15 mm. Similarly, the dimension of the negative terminal assembly **483** is set within the preceding range, which can ensure the connection stability between the negative terminal assembly **483** and the tool terminals **361** and will not cause too long a dimension of the battery pack **40** in the length direction. In this example, the positive terminal assembly **482** and the negative terminal assembly **483** are substantially equal in dimension along the insertion direction of the tool terminals. Further, a communication terminal assembly and the positive terminal assembly **482** are substantially equal in dimension along the insertion direction of the tool terminals. It is to be noted that the dimension F**1** of the positive terminal assembly **482** along the insertion direction of the tool terminals **361** refers to a distance from an end portion of the positive terminal assembly **482** farther from the cell assembly **471** to the interface circuit board **48** along the insertion direction of the tool terminals **361**. Similarly, the dimension F**2** of the negative terminal assembly **483** along the insertion direction of the tool terminals **361** has the same explanation as that of the positive terminal assembly **482**.

(35) Along a direction perpendicular to the first plane P**1**, a projection of the connecting housing portion **421** on the first plane P**1** partially overlaps a projection of the basic housing portion **44** on the first plane P**1**, where a projection area of the overlapping part is less than 90% of a projection area of the basic housing portion **44** on the first plane P**1**, that is, the connecting housing portion **421** only intersects with the basic housing portion **44** partially. Along the direction of the first center line **103**, a ratio of a dimension of the connecting housing portion **421** to a dimension of the basic housing portion **44** is greater than or equal to 0.3 and less than 1, that is, along the direction of the first center line **103**, the dimension of the connecting housing portion **421** is smaller than the dimension of the basic housing portion **44**. Similarly, the dimension of the connecting housing portion **421** is also smaller than a dimension of the extended housing portion **431**. In the direction perpendicular to the first plane P**1**, the base housing section **44**, the connection housing section **421**, and the extension housing section **431** approximately form a recess.

(36) As some examples, an orthographic projection of the gap **60** between the basic housing portion **44** and the extended housing portion **431** on the first plane P**1** may be L-shaped, U-shaped, C-shaped, V-shaped, or the like. The shape of the orthographic projection of the gap **60** between the basic housing portion **44** and the extended housing portion **431** on the first plane P**1** is not limited here as long as the gap **60** can surround the long electrode plate **493** connected between the basic portion **41** and the connecting portion **42**. In this example, the orthographic projection of the gap **60** between the basic housing portion **44** and the extended housing portion **431** on the first plane P**1** is U-shaped, which is not only convenient for manufacturing but also easy for the user to mount the battery pack **40**. Further, the basic housing portion **44** includes a first basic surface **441** and a second basic surface **442** opposite to each other. The extended housing portion **431** includes a first

extension surface **4311** and a second extension surface **4312** opposite to each other. The first basic surface **441** is adjacent to the first extension surface **4311** in the direction perpendicular to the first plane P**1**, and the first basic surface **441** and the first extension surface **4311** are each at least partially connected to the connecting housing portion **421**. A distance H**1** between the first basic surface **441** and the first extension surface **4311** in the direction perpendicular to the first plane P**1** is greater than or equal to 2 mm and less than or equal to 20 mm. It may also be understood as follows: the thickness of the gap **60** is greater than or equal to 2 mm and less than or equal to 20 mm. The thickness of the gap **60** is set within the preceding range. On the one hand, in this manner, it can be ensured that the connection plate **492** and the long electrode plate **493** which are used for connecting the first cell group **45** to the second cell group **46** are not too long so that the electrical connection between the basic housing portion **44** and the extended housing portion **431** is more reliable, and the assembly difficulty of the battery pack **40** is also reduced. On the other hand, in this manner, heat dissipation areas of the basic portion **41** and the extended portion **43** are increased, thereby facilitating the heat dissipation of the battery pack **40**.

(37) A dimension H**2** of the gap **60** in a width direction perpendicular to the first center line **103** and parallel to the first plane P**1** is greater than or equal to 30 mm and less than or equal to 200 mm. The width of the gap **60** is set within the preceding range, which not only can stabilize the connection between the basic housing portion **44** and the extended housing portion **431** but also is convenient for the user to mount the battery pack **40** onto the housing **30**.

(38) To clearly illustrate the technical solutions of the present application, a left side, a right side, an upper side, a lower side, a front side, and a rear side in the present application are shown in FIG. **1** and refer to six orientations of the battery pack **40** in the direction perpendicular to or parallel to the first straight line **101** when the battery pack **40** is mounted onto a battery pack accommodating portion **31**. That is to say, referring to FIGS. **1** and **7**, the first straight line **101** in which the battery pack **40** is coupled to the accommodating portion **31** is the front side, and the side where the first base surface **441** is located is the upper side.

(39) Referring to FIGS. **3** and **4**, the housing **30** is fixedly connected to the battery pack accommodating portion **31**, or the housing **30** and the battery pack accommodating portion **31** are integrally formed. In the present application, the battery pack accommodating portion **31** and the housing **30** are integrally formed. The battery pack accommodating portion **31** includes the first battery pack accommodating cavity **32** and the second battery pack accommodating cavity **33**. The basic portion **41** is at least partially disposed inside the first battery pack accommodating cavity **32**, and the extended portion **43** is at least partially disposed inside the second battery pack accommodating cavity **33** when the battery pack **40** is mounted to the battery pack accommodating portion **31**. That is, the first cell group **45** is at least partially disposed inside the first battery pack accommodating cavity **32**, and the second cell group **46** is at least partially disposed outside the first battery pack accommodating cavity **32**. In this example, the second cell group **46** is at least partially disposed inside the second battery pack accommodating cavity **33**. In this manner, the battery packs **40** with different capacities can be mounted to the battery pack accommodating portion **31** according to requirements of the user, which offers more choices to the user and is more convenient for the user. In addition, the capacity of the extended portion **43** of the battery pack **40** can be extended without limit according to the requirements of the user, which can effectively improve the endurance of the blower **100**, prolong the working time of the blower **100**, and facilitate operation by the user because the user does not need to frequently replace the battery pack **40**. In the present application, the second battery pack accommodating cavity **33** may be understood as an external space.

(40) The first battery pack accommodating cavity **32** includes a first sidewall **321**, a second sidewall **322**, a third sidewall **323**, a fourth sidewall **324**, and a bottom wall **325**. That is, the first battery pack accommodating cavity **32** is surrounded and formed by the preceding four sidewalls and bottom wall **325**. Specifically, when the battery pack **40** is mounted to the battery pack

accommodating portion **31**, the first sidewall **321** is disposed on a right side of the battery pack **40**; the second sidewall **322** is disposed opposite to the first sidewall **321**, that is, the second sidewall **322** is disposed on a left side of the battery pack **40**; the third sidewall **323** is disposed on an upper side of the battery pack **40**; and the fourth sidewall **324** is disposed opposite to the third sidewall **323**, that is, the fourth sidewall **324** is disposed on a lower side of the battery pack **40**. An outer wall of the fourth sidewall **324** defines a limit plane A parallel to the first straight line **101**, where the basic portion **41** and the extended portion **43** of the battery pack **40** are disposed on two sides of the limit plane A, separately. Further, the basic portion **41** is disposed inside the first battery pack accommodating cavity **32**, and the extended portion **43** is disposed outside the first battery pack accommodating cavity **32**. It may be understood as follows: the basic portion **41** is disposed on a side of the fourth sidewall **324** facing towards the third sidewall **323**, and the extended portion **43** is disposed on a side of the fourth sidewall **324** facing away from the third sidewall **323**. In this manner, the battery packs **40** with different capacities can be connected without completely resetting the battery pack accommodating portion **31**. Additionally, in this manner, the battery capacity of the extended portion **43** outside the first battery pack accommodating cavity **32** may not be limited by a structure so that the capacity of the extended portion **43** can be extended without limit, thereby improving the performance and endurance of the blower **100**. It is to be noted that the sidewalls here are not limited to planes. In addition, it is to be noted that the fourth sidewall **324** may not be limited to only one component. A part of the fourth sidewall **324** may be fixedly connected to the first sidewall **321** or may be directly formed by the first sidewall **321** extending into the first battery pack accommodating cavity **32**. The other part of the fourth sidewall **324** may be fixedly connected to the second sidewall **322** or may be directly formed by the second sidewall **322** extending into the first battery pack accommodating cavity **32**. That is, the fourth sidewall **324** may be constituted by two components. The lengths of the part and the other part of the fourth sidewall **324** in a direction which is parallel to or coincides with the limit plane A are not limited here, that is, the length of the fourth sidewall **324** may be equal to 0 mm, that is, there is no third housing portion **313**. In this manner, the first battery pack accommodating cavity **32** formed by the first sidewall **321**, the second sidewall **322**, the third sidewall **323**, and the bottom wall **325** is substantially semi-enclosed and U-shaped. As another example, of course, the fourth sidewall **324** may also be constituted by multiple components. In this example, the fourth sidewall **324** is one component.

(41) In this example, the housing **30** includes a first housing portion **311**, a second housing portion **312**, and a third housing portion **313**, where the first housing portion **311** forms the first sidewall **321**, the second housing portion **312** forms the second sidewall **322**, the third sidewall **323** and the bottom wall **325** are each formed by part of the first housing portion **311** and part of the second housing portion **312** together, and the fourth sidewall **324** is formed by the third housing portion **313**. It may also be understood as follows: the first sidewall **321** and a part of the third sidewall **323** are integrally formed as a first integral piece **326**, and the second sidewall **322** and the other part of the third sidewall **323** are integrally formed as a second integral piece **327**, where the first integral piece **326** is detachably connected to the second integral piece **327**, the fourth sidewall **324** connects the first integral piece **326** to the second integral piece **327**, and the fourth sidewall **324** is detachably connected to the first integral piece **326** and the second integral piece **327**. It is to be noted that the detachable connection refers to causing a component to be detached from an original structure without damaging the original structure, and the detachable connection here refers to that the first integral piece **326** and the second integral piece **327** are detached from each other without damaging their original structures. That is, a snap-fit connection, a threaded connection, a connection through screws **315**, or the like may be used between the first housing portion **311** and the second housing portion **312** as long as the first housing portion **311** and the second housing portion **312** are not separated from each other during normal working. In the example, the first housing portion **311** and the second housing portion **312** are connected through the screws **315**.

Similarly, the snap-fit connection, the threaded connection, the connection through screws **315**, or the like may be used between the fourth sidewall **324** and the first integral piece **326** and/or the second integral piece **327**. In this example, the fourth sidewall **324** is detachably connected to the second sidewall **322**. Specifically, the fourth sidewall **324** is connected to the second sidewall **322** through the screws **315**. To ensure the connection stability, one side of the fourth sidewall **324** is detachably connected to the second sidewall **322**, and the other side of the fourth sidewall **324** is detachably connected to the first sidewall **321**. That is, the fourth sidewall **324** is fixed to the second sidewall **322** through the screws **315**, and the fourth sidewall **324** is fixed to the first sidewall **321** through the screws **315**. Thus, after the battery pack **40** is mounted to the battery pack accommodating portion **31**, the battery pack **40** is prevented from being separated from the battery pack accommodating portion **31** because of shakes when the motor **20** is working. Moreover, when a component inside the first battery pack accommodating cavity **32** fails, the fourth sidewall **324** can be easily removed by the user, thereby facilitating maintenance by the user. Further, to improve the structural strength of the first battery pack accommodating cavity **32**, reinforcement ribs may be disposed on the sidewalls.
(42) Referring to FIG. **5** as well, the blower **100** further includes a midplane B perpendicular to the limit plane A. The fourth sidewall **324** is symmetrical about the midplane B. In this example, the first sidewall **321** is disposed on a left side of the midplane B, and the second sidewall **322** is disposed on a right side of the midplane B. It is to be noted that the fourth sidewall **324** being symmetrical about the midplane B in the present application is not strictly limited to the fourth sidewall **324** being completely identical on the left and right sides of the midplane B. It may be understood as the fourth sidewall **324** being symmetrical about the midplane B as long as an error is within an allowable range. The fourth sidewall **324** is disposed between the first battery pack accommodating cavity **32** and the second battery pack accommodating cavity **33**, where the fourth sidewall **324** is provided with an opening **35** through which the first battery pack accommodating cavity **32** communicates with the second battery pack accommodating cavity **33**, and the midplane B passes through the preceding opening **35**. It may also be understood as follows: the opening **35** penetrates through the battery pack accommodating portion **31**. That is, the opening **35** formed on the fourth sidewall **324** penetrates through the first battery pack accommodating cavity **32** and the second battery pack accommodating cavity **33** along a direction of a second straight line **102**, and the opening **35** opens toward a direction away from the bottom wall **325** along the direction of the first straight line **101**. When the battery pack **40** is mounted to the battery pack accommodating portion **31**, the connecting housing portion **421** for connecting the extended portion **43** to the basic portion **41** passes through the preceding opening **35**, that is, the opening **35** guides the battery pack **40** to be coupled to the first battery pack accommodating cavity **32**. With the preceding arrangement, it is convenient for the user to mount the battery pack **40** into the battery pack accommodating portion **31**. In the present application, the first straight line **101** is perpendicular to the second straight line **102**.
(43) Further, a ratio of a dimension of the opening **35** along the direction of the first straight line **101** to a dimension of the first battery pack accommodating cavity **32** along the direction of the first straight line **101** is less than 1. It is to be noted that the dimension of the first battery pack accommodating cavity **32** along the direction of the first straight line **101** refers to a dimension from the bottom wall **325** to an end of the sidewall farthest from the bottom wall **325** along the direction of the first straight line **101**. A distance of the opening **35** along the direction of the first straight line **101** refers to a dimension from the end of the sidewall farthest from the bottom wall **325** along the direction of the first straight line **101** to an end of the opening **35** closest to the bottom wall **325** along the direction of the first straight line **101**. Specifically, the ratio of the dimension L**1** of the opening **35** along the direction of the first straight line **101** to the dimension L**2** of the first battery pack accommodating cavity **32** along the direction of the first straight line **101** is greater than or equal to 0.3 and less than 1. In this example, the opening **35** can guide the

connecting housing portion **421** of the battery pack **40** to be slidably inserted into the first battery pack accommodating cavity **32** along the direction of the first straight line **101**. That is, in a direction perpendicular to the first straight line **101** and parallel to the limit plane A, a dimension of the opening **35** should be greater than a dimension of the connecting housing portion **421**, and the connecting housing portion **421** should be adjacent to the opening **35** when the battery pack **40** is mounted onto the accommodating cavity of the battery pack **40**. The dimension of the opening **35** and a dimension of the first accommodating cavity **411** are set within the preceding ranges so that an enough space is provided for the connecting housing portion **421** of the battery pack **40** to move.
(44) In some other examples, the ratio of the dimension L**1** of the opening **35** along the direction of the first straight line **101** to the dimension of the first battery pack accommodating cavity **32** along the direction of the first straight line **101** is greater than or equal to 0.4 and less than or equal to 0.9. In some other examples, the ratio of the dimension of the opening **35** along the direction of the first straight line **101** to the dimension of the first battery pack accommodating cavity **32** along the direction of the first straight line **101** is greater than or equal to 0.6 and less than or equal to 0.8. The connecting housing portion **421** is adjacent to the opening **35**, the ratio of the dimension of the opening **35** along the direction of the first straight line **101** to the dimension of the first battery pack accommodating cavity **32** along the direction of the first straight line **101** is set within the preceding range, and it may also be understood as follows: a distance from the connecting housing portion **421** to the bottom wall **325** of the first battery pack accommodating cavity **32** is substantially within the preceding range. In this manner, the connection between the basic portion **41** and the extended portion **43** can be more stable. In addition, since the connecting housing portion **421** for connecting the basic portion **41** to the extended portion **43** is disposed in the opening **35**, the battery pack **40** can be prevented from arbitrarily swaying in the accommodating cavity of the battery pack **40** even in the case of vigorous shakes.
(45) As shown in FIGS. **2** to **7**, the blower **100** further includes a locking assembly **50** disposed on the third sidewall **323**. The locking assembly **50** is used for locking the battery pack **40** into the first battery pack accommodating cavity **32**. It may also be understood as follows: the locking assembly **50** includes at least a first state (as shown in FIG. **2**) and a second state (as shown in FIG. **3**). When the locking assembly **50** is in the first state, the battery pack **40** remains fixed relative to the first battery pack accommodating cavity **32**. When the locking assembly **50** is in the second state, the battery pack **40** can be separated from the first battery pack accommodating cavity **32**.
(46) The locking assembly includes an operation member **51**, a biasing member **52**, and a connection shaft **53**. The operation member **51** can be driven to rotate about a first axis **105**, where the operation member **51** includes at least a locking position (as shown in FIG. **2**), a release position (as shown in FIG. **3**), and an initial position (as shown in FIG. **4**) during rotation. The initial position refers to a position where the operation member **51** without being driven by an external force is located when the battery pack **40** is not mounted onto the housing **30**. When the operation member **51** is rotated about the first axis **105** along a first direction **1051** from the release position to the locking position, the battery pack **40** is fixed relative to the housing **30**, that is, the locking assembly **50** is in the first state. When the operation member **51** is rotated about the first axis **105** along a second direction **1052** from the locking position to the release position, the battery pack **40** can move relative to the housing **30**, that is, the locking assembly **50** is in the second state. The biasing member **52** is used for biasing the operation member **51**, that is, the biasing member **52** is mounted with a biasing force. Thus, the operation member **51** is rotated about the first axis **105** along the first direction **1051** without the influence of the external force. That is, without the external force, the operation member **51** will be in the initial position due to the biasing force. When the user mounts the battery pack **40** onto the tool body **100***a*, the user needs to apply a force against the preceding biasing force to rotate the operation member **51** to the release position. The connection shaft **53** is mounted on the housing **30**, and both the operation member **51** and the biasing member **52** are sleeved on the connection shaft **53**. An operation portion **511**, a locking

portion **512**, and a middle portion **513** for connecting the operation portion **511** to the locking portion **512** are fixedly connected to or integrally formed on the operation member **51**. The operation portion **511** and the locking portion **512** are disposed at two ends of the operation member **51** and move synchronously with the operation member **51**. The operation portion **511** is operated by the user, that is, the user rotates the operation member **51** by driving the operation portion **511**. The locking portion **512** mates with the battery pack **40** to implement locking and release. Further, a slot portion **412** disposed on an upper surface of the battery pack **40** can be engaged with the locking portion **512**, thereby implementing locking. The battery pack **40** is released when the slot portion **412** is separated from the locking portion **512**. In this example, the operation portion **511**, the locking portion **512**, and the operation member **51** are integrally formed. In a direction perpendicular to the limit plane A, a cross section of the operation member **51** is substantially Z-shaped. That is, in the direction perpendicular to the limit plane A, the operation portion **511** of the operation member **51** is higher than the locking portion **512**, that is, the middle portion **513** connects the operation portion **511** to the locking portion **512** in an inclined manner. That is, in the direction perpendicular to the limit plane A, a bypass space **314** is formed between the operation portion **511**, the first housing portion **311**, and the second housing portion **312** for disposing the screws **315** fixedly connecting the first housing portion **311** to the second housing portion **312**.

(47) The tool body **100***a* further includes a terminal block **36** on the bottom wall **325** and a resilient assembly **90** adjacent to the terminal block **36**. The terminal block **36** is fixedly connected to the housing **30**, and the tool terminals **361** and a positioning column **362** are fixedly connected to the terminal block **36**. The tool terminals **361** are used for connecting the battery pack **40** so that the battery pack **40** powers the tool body **100***a*. Specifically, the tool terminals **361** are electrically connected to the electrode plate terminal assembly **481** of the battery pack **40**. The positioning column **362** is fixedly connected to the terminal block **36**, or the positioning column **362** and the terminal block **36** are integrally formed. The positioning column **362** is used for guiding the battery pack **40** to be mounted to the locking position along the direction of the first straight line **101**. Two or more tool terminals **361** extending along the direction of the first straight line **101** are at least disposed on the terminal block **36**, that is, an extension direction of the positioning column **362** is substantially consistent with an extension direction of the tool terminals **361**. Along the direction of the first straight line **101**, a length of a portion of the tool terminal **361** protruding from the terminal block **36** is greater than a length of a portion of the positioning column **362** protruding from the terminal block **36**, that is, the tool terminal **361** is higher than the positioning column **362**. With the preceding arrangement, when the battery pack **40** is slidably mounted to the locking position along the direction of the straight line **101**, the tool terminals **361** are firstly inserted into the electrode plate terminal assembly **481** of the battery pack **40**. That is, the tool terminals **361** are aligned with an electrode plate terminal assembly **481**, and then the battery pack **40** is finally mounted to the locking position by being guided by the positioning column **362**. The hardness of both the tool terminals **361** and the electrode plate terminal assembly **481** of the battery pack **40** is less than the hardness of the positioning column **362**, which is convenient for the user to better insert the tool terminals **361** into the electrode plate terminal assembly **481**, thereby avoiding ignition caused by the misalignment between the tool terminals **361** and the electrode plate terminal assembly **481** after the positioning column **362** is aligned. In addition, the positioning column **362** has greater hardness than the tool terminals **361**, which can ensure the connection stability between the battery pack **40** and the terminal block **36** and avoid the separation between terminals or the deformation of terminals due to shakes when the tool is working.

(48) The resilient assembly **90** resiliently abuts against the battery pack **40**. On the one hand, the resilient assembly **90** is used for supporting the battery pack **40**. On the other hand, when the battery pack **40** is released from the battery pack accommodating portion **31**, the resilient assembly **90** applies a force on the battery pack **40** to move the battery pack **40** to the rear side, thereby

ejecting the battery pack **40**. The resilient assembly **90** includes a support element **91** and a resilient element **92**. The support element **91** is used for abutting against the battery pack housing **47** when the battery pack **40** is mounted to the locking position. The resilient element **92** is used for abutting against the support element **91**. Specifically, the support element **91** is provided with a support portion protruding toward the first battery pack accommodating cavity **32** substantially along the direction of the first straight line **101**. The resilient element **92** is at least partially disposed inside the support element **91**. One end of the resilient element **92** is fixed onto the housing **30**, and the other end of the resilient element **92** abuts against the support element **91**. When the battery pack **40** is in the locking position, along the direction of the first straight line **101**, a length of a portion of the support element **91** protruding from the terminal block **36** is less than the length of the portion of the positioning column **362** protruding from the terminal block **36**, and a distance from the support element **91** to the bottom wall **325** is greater than 0 mm. That is, when the battery pack **40** is mounted in place, the battery pack housing **47** is in contact with the support element **91**, and an interval exists between the battery pack housing **47** and the terminal block **36**. When the user selects a large-capacity battery pack **40** as required, since the battery pack **40** is heavy, the battery pack housing **47** easily damages the terminal block **36** when the battery pack **40** is mounted to the first battery pack accommodating cavity **32**. With the preceding arrangement, the impact force brought by the battery pack **40** is buffered with the property of the resilient element **92**, so as to prevent the terminal block **36** from being damaged and prolong the service life of the terminal block **36**. In addition, when the battery pack **40** is released, a deformation force generated by the resilient element **92** due to the battery pack **40** moves the battery pack **40** along a direction opposite to the insertion direction, thereby facilitating the removal of the battery pack **40** by the user.

(49) The battery pack accommodating portion **31** is configured for the battery pack **40** to be at least partially inserted into along the direction of the first straight line **101**. It may be understood as follows: the battery pack **40** may be inserted into the battery pack accommodating portion **31** along the direction of the first straight line **101**. In this example, a guide structure **34** is disposed on the battery pack accommodating portion **31** for guiding the battery pack **40** to be inserted into the battery pack accommodating portion **31** along the direction of the first straight line **101**. The guide structure **34** is disposed inside the first battery pack accommodating cavity **32** and adjacent to end portions of the sidewalls. Specifically, the guide structure **34** is disposed on the sidewalls of the first battery pack accommodating cavity **32**. The guide structure **34** includes at least a pair of opposite guide ribs **341**. The guide ribs **341** are defined here as a first guide rib **3411** and a second guide rib **3412**. The first guide rib **3411** is fixedly connected to the first sidewall **321**, or the first guide rib **3411** and the first sidewall **321** are integrally formed. The second guide rib **3412** is fixedly connected to the second sidewall **322**, or the second guide rib **3412** and the second sidewall **322** are integrally formed. In this example, the guide ribs **341** and the sidewalls are integrally formed, that is, the first guide rib **3411** protruding toward the first battery pack accommodating cavity **32** is formed on the first sidewall **321**, and the second guide rib **3412** protruding toward the first battery pack accommodating cavity **32** is formed on the second sidewall **322**. The guide ribs **341** are disposed on the first sidewall **321** and the second sidewall **322**, which not only facilitates the mounting of the battery pack **40** but also positions the battery pack **40** on the left and right sides, thereby preventing the battery pack **40** from swaying in the first battery pack accommodating cavity **32**. The first guide rib **3411** and the second guide rib **3412** include multiple discontinuous positioning protrusions. Guide grooves **413** corresponding to the guide ribs **341** are formed on the basic housing portion **44** of the battery pack **40**. The battery pack housing **47** is recessed inward along the length direction of the battery pack **40** so as to form the guide grooves **413**. The guide grooves **413** on the battery pack **40** mate with the positioning protrusions to position the battery pack **40**. Multiple protrusions are disposed on the guide ribs **341** so that positioning problems caused by the deformation of the guide ribs **341** can be effectively avoided. It is to be noted that a length by which the guide rib **341** extends substantially along the direction of the first straight line

101 is not limited here.
(50) Further, opposite guide ribs **341** are fixedly connected to the fourth sidewall **324** and the third sidewall **323**, or the opposite guide ribs **341** are integrally formed on the fourth sidewall **324** and the third sidewall **323**, so as to position the battery pack **323** on the upper and lower sides. Specifically, a third guide rib **3413** is disposed on the third sidewall **323**, and a fourth guide rib **3414** is disposed on the fourth sidewall **324**. The fourth guide rib **3414** is formed on the fourth sidewall **324**, and the third guide rib **3413** is fixedly connected to the third sidewall **323**. That is, the third guide rib **3413** may be separated from the third sidewall **323**, and the third guide rib **3413** is a separate component, which facilitates the demoulding of the first housing portion **311** and the second housing portion **312**.
(51) The above illustrates and describes basic principles, main features, and advantages of the present disclosure. It is to be understood by those skilled in the art that the preceding examples do not limit the present disclosure in any form, and technical solutions obtained by means of equivalent substitutions or equivalent transformations fall within the scope of the present disclosure.

Claims

1. A power tool, comprising: an output member for outputting power; a motor for driving the output member; a housing; and a battery pack for powering the motor, wherein the housing comprises: a battery pack accommodating portion provided with a first battery pack accommodating cavity and a guide structure for guiding the battery pack to be inserted into the first battery pack accommodating cavity along a first straight line, wherein the battery pack comprises: a basic housing portion provided with a first accommodating cavity; a first cell group disposed inside the first accommodating cavity; an extended housing portion provided with a second accommodating cavity; and a second cell group disposed inside the second accommodating cavity, and wherein the first cell group is at least partially disposed inside the first battery pack accommodating cavity and the second cell group is at least partially disposed outside the first battery pack accommodating cavity.
2. The power tool according to claim 1, wherein the first cell group comprises a plurality of first cells, the second cell group comprises a plurality of second cells, the plurality of first cells are disposed inside the first accommodating cavity, the plurality of second cells are disposed inside the second accommodating cavity, and a ratio of a number of the plurality of first cells to a number of the plurality of second cells is greater than or equal to 0.3 and less than or equal to 2.
3. The power tool according to claim 2, wherein the battery pack accommodating portion is further provided with the second battery pack accommodating cavity, the basic housing portion is disposed inside the first battery pack accommodating cavity, and the extended housing portion is disposed inside the second battery pack accommodating cavity.
4. The power tool according to claim 2, wherein each of the plurality of first cells is a cylinder centered on a first center line, each of the plurality of second cells is a cylinder centered on a second center line, and the first center line and the second center line are parallel to each other.
5. The power tool according to claim 2, wherein each of the plurality of first cells is a cylinder centered on a first center line, each of the plurality of second cells is a cylinder centered on a second center line, and the first center line and the second center line are perpendicular to each other.
6. The power tool according to claim 1, wherein a projection of the first cell group in a plane perpendicular to the first straight line does not overlap a projection of the second cell group in the plane.
7. The power tool according to claim 1, wherein the first battery pack accommodating cavity comprises a first sidewall disposed on a right side of the battery pack, a second sidewall opposite to

the first sidewall, a third sidewall disposed on an upper side of the battery pack, and a fourth sidewall opposite to the third sidewall, an outer wall of the fourth sidewall defines a limit plane parallel to the first straight line, and the battery pack comprises a basic portion on one side of the limit plane and an extended portion on another side of the limit plane.

8. The power tool according to claim 7, wherein the fourth sidewall is provided with an opening penetrating through the battery pack accommodating portion along a second straight line, the opening opens rearward along the first straight line, and the second straight line is perpendicular to the first straight line.

9. The power tool according to claim 8, wherein a housing of the battery pack comprises a connecting housing portion for connecting the extended portion to the basic portion and the connecting housing portion passes through the opening.

10. The power tool according to claim 8, wherein a ratio of a dimension of the opening along the first straight line to a dimension of the first battery pack accommodating cavity along the first straight line is greater than or equal to 0.3 and less than 1.

11. The power tool according to claim 8, wherein a ratio of a dimension of the opening along the first straight line to a dimension of the first battery pack accommodating cavity along the first straight line is greater than or equal to 0.6 and less than or equal to 0.8.

12. The power tool according to claim 8, wherein the first sidewall is disposed on a right side of a midplane, the second sidewall is disposed on a left side of the midplane, and the midplane passes through the opening.

13. The power tool according to claim 7, wherein the first sidewall and a part of the third sidewall are integrally formed as a first integral piece, the second sidewall and another part of the third sidewall are integrally formed as a second integral piece, the first integral piece is detachably connected to the second integral piece, and the fourth sidewall connects the first integral piece to the second integral piece.

14. The power tool according to claim 7, wherein the power tool has a midplane perpendicular to the limit plane and the fourth sidewall is symmetrical about the midplane.

15. The power tool according to claim 1, wherein the battery pack further comprises a connecting housing portion for connecting the basic housing portion to the extended housing portion and the connecting housing portion is disposed between the basic housing portion and the extended housing portion such that a gap outside a housing of the battery pack is formed between the basic housing portion and the extended housing portion.

16. The power tool according to claim 15, wherein a thickness of the gap is greater than or equal to 2 mm and less than or equal to 20 mm.

17. The power tool according to claim 15, wherein the first cell group comprises a plurality of first cells each being a cylinder centered on a first center line, three or more first cells are disposed in a first plane parallel to the first center line, and a projection of the gap in the first plane is U-shaped.